

Yes, there are several common metrics and models for calculating hardware lifespan reduction due to heat:

1. The Arrhenius Equation

The Arrhenius equation is the fundamental formula used to predict the rate of failure or degradation of components based on temperature. The equation is:

$$AF = \exp[(E_a/k) \times (1/T_{\text{use}} - 1/T_{\text{test}})]$$

Where:

- AF = Acceleration Factor (how much faster degradation occurs)
- E_a = Activation energy (specific to the failure mechanism)
- k = Boltzmann constant (8.62×10^{-5} eV/K)
- T = Absolute temperature in Kelvin

The acceleration factor relates the life of a component at its normal use temperature to test time at elevated temperature.

2. The "10°C Rule" (Rule of Thumb)

The most common rule of thumb states that every 10°C increase in temperature reduces component life by half. However, this is an **oversimplification**.

For this rule to be accurate, the activation energy needs to be in a specific range, and it typically applies for activation energies around 0.6-1.0 eV/K in the temperature range of 75-125°C.

Important Caveats:

The Arrhenius model is appropriate for certain failure mechanisms including corrosion, electromigration, and certain manufacturing defects, but is not suitable for other significant failure modes such as formation of conductive filaments, contact interface stress relaxation, and fatigue of package-to-board level interconnects.

As an example, testing on a commercial resistive switching memory device found an activation energy of 1.13 eV/K, where increasing operating temperature from 45°C to 55°C would give an acceleration factor of approximately 3.8 - not 2.

3. Typical Activation Energy Values

Many electronics failure mechanisms have reported activation energies in the range of approximately 0.6-1.0 eV/K.

Practical Example:

For a product with a normal operating temperature of 50°C tested at 100°C with an activation energy of 0.7 eV, the acceleration factor would be 29. This means testing for 1,000 hours (about 6 weeks) at 100°C is equivalent to 29,000 hours or 3.3 years at 50°C.

For semiconductors, if a processor runs at 90°C effective temperature instead of 105°C, a 2x increase in useful lifetime can be projected, potentially achieving a 20-year useful lifetime if thermal performance is managed to 90°C or below.

Bottom Line:

While the "10°C = half-life" rule is widely used, it's **only a rough approximation** that works for specific failure mechanisms and temperature ranges. For accurate predictions, you need to know the specific activation energy for your component and failure mode, then apply the Arrhenius equation properly.